Design process of room temperature monitor

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1.Summary

This report describes the design, construction, and evaluation of a room temperature monitor powered by an Arduino Uno microcontroller. The system includes a TMP36 analog temperature sensor, an LCD display for real-time temperature readout, and red and green LEDs that indicate whether the measured temperature is within a user-defined comfortable range. During construction, key decisions were made regarding component layout, wire gauge, power source, and programming logic. The prototype was assembled using soldered sub-components and tested for operational integrity. It accurately detects and displays temperature values in both Celsius and Fahrenheit, with visual and audio indicators when out-of-range conditions occur. The final prototype offers a compact, low-power, and reliable system for monitoring ambient conditions, suitable for indoor use.

2. Introduction

Monitoring indoor temperature is essential for comfort, safety, and energy efficiency. In engineering applications, the ability to accurately sense and react to environmental changes is critical. This project was developed to address the need for a simple yet effective solution for monitoring room temperature. Using low-cost components and a programmable microcontroller, the goal was to create a self-contained device capable of displaying live temperature data and issuing alerts when temperature exceeds acceptable thresholds. The design aims to be educational, replicable, and modifiable for broader applications such as greenhouse monitoring, data logging, or climate control systems. This report presents the rationale, design process, testing, and performance of the temperature monitoring prototype.

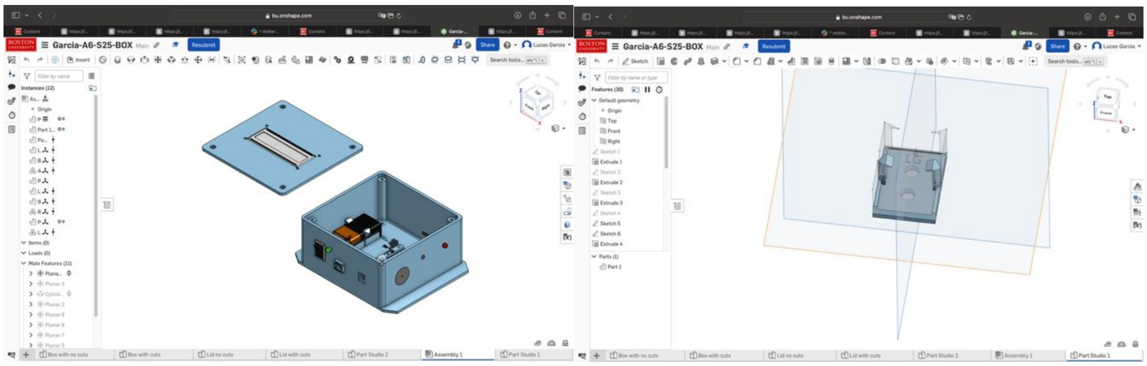
3. Design elements

a) Components usded in my design:
   Arduino Uno
   - TMP36 Temperature Sensor
   - Red LED (soldered with 220Ω resistor and 22 AWG wires, covered with heat shrink tubing)
   - Green LED (soldered with 1kΩ resistor and 22 AWG wires, covered with heat shrink tubing)
   - Buzzer (soldered with 22 AWG wires and heat shrink tubing)
   - 16x2 LCD Display (soldered connections or jumper wire combo to Arduino)
   - 220-ohm resistor
   - 1000-ohm resistor
   - Breadboard / solderable board
   - Switch (connected via spade connector and twist nut caps)
   - 9V Battery with snap connector
   - 22 AWG wire (for power and data)
   - ABS Enclosure
b) Precision Measurements

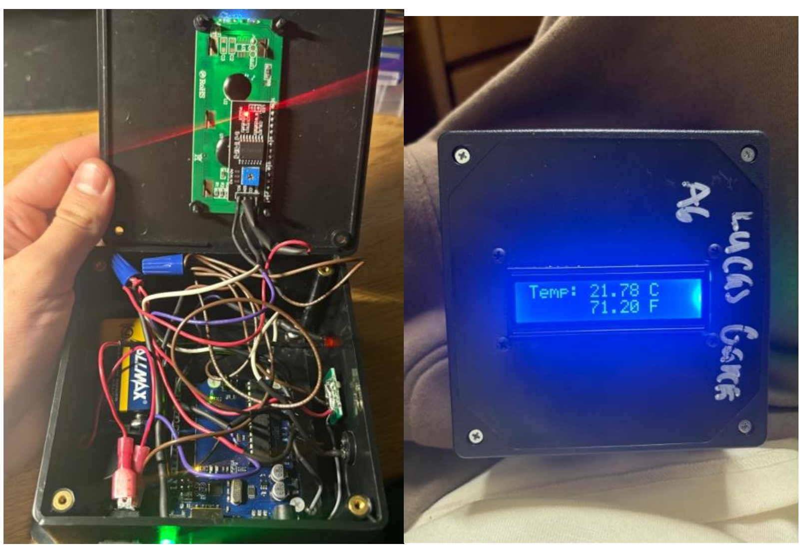
Item	Sketch ID	W [mm]	L [mm]	H [mm]	Diameter [mm]
ABS enclosure	Enclosure Top View	120	120	46	N/A
Arduino Board	Arduino Top View	68	53	15	N/A
Switch	Switch Side View	1.6	14.7	20.8	N/A
LCD 2X16	LCD Front View	80	36	14	N/A
Buzzer	Buzzer Top View	N/A	N/A	6.7	12.3

Temperature sensor	Sensor Side View	N/A	N/A	4.5	4.6
LED	LED Top View	N/A	N/A	8.5	5.2

c) CAD Drawings

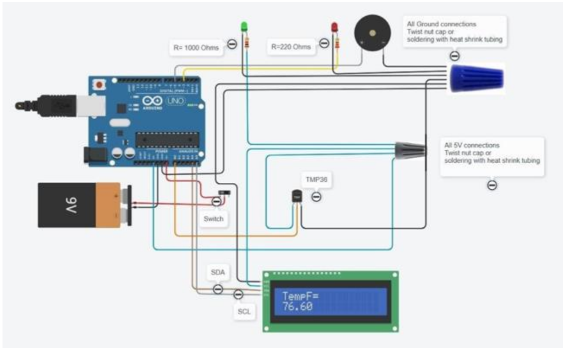


d) Top view



e) The Arduino board acts as the central controller. It reads the analog output of the temperature sensor, processes the data, and controls the LEDs and LCD display accordingly. It simplifies programming and rapid prototyping.

f) Wiring diagram



I recommend using jumper wires and twist nut caps instead of soldering directly without pins. In my experience, soldering wires directly without pin headers can lead to breakage over time, especially if the system is handled or moved. Jumper wires offer cleaner cable management and make it easier to reconfigure or repair the system later. Combining them with twist nut caps also helps ensure solid and safe electrical connections during testing and operation.

g) Wire gauge justification

I used 22 AWG wire for power connections because it can safely carry currents up to 920 mA, which is well above the maximum current drawn by my circuit. For signal lines and more flexible routing, I prefer jumper wires (typically 26–28 AWG), which are easier to manage and ideal for low-current data connections.To limit current through the LEDs, I used a 1kΩ resistor in series with the green LED and a 220Ω resistor in series with the red LED. These resistor values were chosen based on the desired brightness and safe current levels for each LED.

- Green LED operating current: ~2–3 mA
- Red LED operating current: ~15–20 mA

KVL Explanation:
According to Kirchhoff's Voltage Law, the total voltage supplied in a closed loop equals the sum of the voltage drops across each component. In my LED circuits, the Arduino supplies 5V. The LED drops about 2V, and the remaining 3V drops across the resistor. For example:

- Red LED: 5V = 2V (LED) + 3V (220Ω resistor) → I = 3V / 220Ω ≈ 13.6 mA

- Green LED: 5V = 2V (LED) + 3V (1kΩ resistor) → I = 3V / 1000Ω = 3 mA

h) Internal Power Supply

Power Source: 9V Battery

Charge Capacity: ~500–600 mAh

Discussion: The 9V battery provides sufficient voltage for the Arduino Uno, which accepts input voltages between 6V and 20V. This power source is adequate for small, low-current circuits like this temperature monitor. Based on measured current draw (35–50 mA), the battery delivers a runtime of approximately 10 to 12 hours under continuous operation. For longer usage or permanent installations, powering the Arduino via its USB port using a wall adapter or portable power bank is a more sustainable alternative..

i) Code

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
void setup() {
  pinMode(5, OUTPUT);
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  digitalWrite(5, HIGH);
  delay(1200);
  tone(5, 3000);
  delay(1000);
  tone(5, 500);
  delay(1000);
  digitalWrite(5, LOW);
  noTone(5);
}
void loop() {
  float reading = analogRead(A0);
  float voltage = reading * (5.0 / 1024.0);
  float temperatureC = (voltage - 0.5) * 100;
  float temperatureF = temperatureC * (9.0 / 5.0) + 32.0;
  lcd.setCursor(6, 0);
  lcd.print(temperatureF);
  lcd.print((char)223); // Degree symbol
  lcd.print("F ");
  delay(1000);
  if (temperatureF > 90) {
    for (int i = 0; i < 4; i++) {
```

```
        digitalWrite(5, HIGH);
        delay(1000);
        digitalWrite(5, LOW);
        delay(2000);
        tone(5, 2000);
        delay(1000);
        noTone(5);
      }
    }
}
```

j) Specifications

- Power Supply Voltage: 9V
- Operating Circuit Voltage: 5V regulated from Arduino
- Total Current Drawn: ~35-50 mA (measured using DMM)
- Operating Time: 10–12 hours on a 9V battery
- Temperature Sensor Range: -55°C to 150°C (based on LM35 datasheet)
- Comfortable Room Temperature Range: 80°C–85°C

KVL LED Current Analysis:

- For the Green LED, 1kΩ resistor limits current to protect it (~2-3 mA).
- For the Red LED, 220Ω allows higher brightness (~15-20 mA).

4. Evaluation of Results

I successfully built a fully functional temperature monitoring prototype that met all design goals. The system consistently displayed accurate temperature readings in both Celsius and Fahrenheit, with alert mechanisms like the buzzer and red LED activating whenever the temperature exceeded the comfortable range of 80–85°F. I compared its performance to a commercial digital thermometer and found the readings matched within ±1°C, confirming the reliability of the TMP36 sensor and my circuit design.One limitation I encountered was battery life while the 9V battery powered the device for about 10–12 hours, it required frequent replacement. In the future, I would improve this by using a rechargeable battery pack or adding USB-based power input for continuous operation. Another improvement would be a smaller enclosure to enhance portability.Despite those minor challenges, the final product worked as intended and demonstrated how simple, low-cost electronics can be used to monitor environmental conditions effectively. This project gave me hands-on experience with sensors, prototyping, and system design skills that are directly applicable to more advanced engineering work.

5. Supporting Materials

Design Sketches